

Henderson-Hasselbalch Model

Acetate-buffer relation

$$\text{pH} = \text{p}K_a + \log_{10} \frac{[A^-]}{[HA]}$$

$$C_{HA}^{\text{mix}} = \frac{C_0 F_H}{F_H + F_A + F_W}, \quad C_{A^-}^{\text{mix}} = \frac{C_0 F_A}{F_H + F_A + F_W}$$

Why water cancels

$$\frac{C_{A^-}^{\text{mix}}}{C_{HA}^{\text{mix}}} = \frac{C_0 F_A / (F_H + F_A + F_W)}{C_0 F_H / (F_H + F_A + F_W)} = \frac{F_A}{F_H}$$

- The same total-flow denominator appears in both buffer components.
- With equal 100 mM stocks, the steady-state pH ratio is acid/acetate only.

Steady-state calculation

$$\text{pH}_{HH,k} = \text{p}K_a + \log_{10} \left(\frac{F_{\text{acetate},k}}{F_{\text{acid},k}} \right)$$

- Water is not a primary steady-state pH actuator.
- It dilutes acid and acetate together.
- The acid/acetate ratio is unchanged.

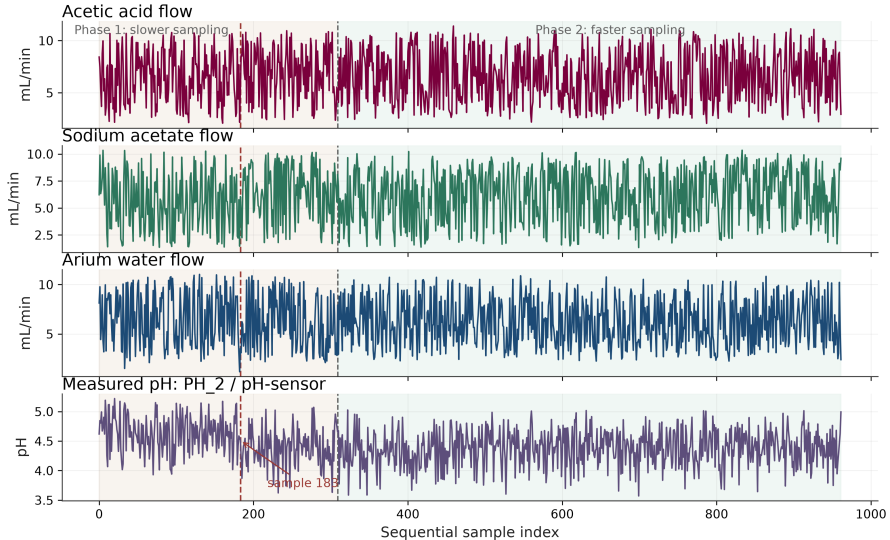
Where water still matters

$$F_T = F_H + F_A + F_W, \quad \tau \approx \frac{V}{F_T}$$

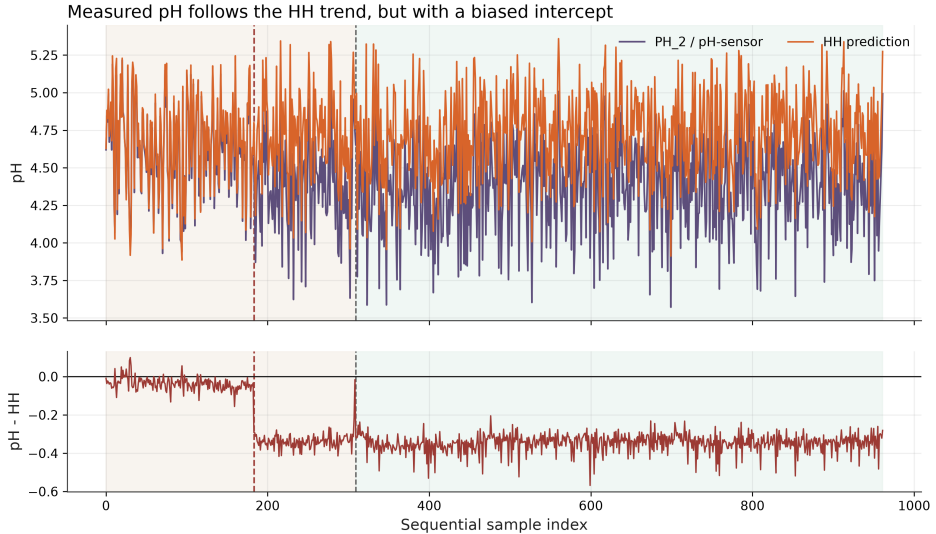
$$C_{\text{buf}} = \frac{C_0 (F_H + F_A)}{F_H + F_A + F_W}$$

- dilution and buffer strength,
- hydraulic residence time and transport delay,
- mixing and sensor response.

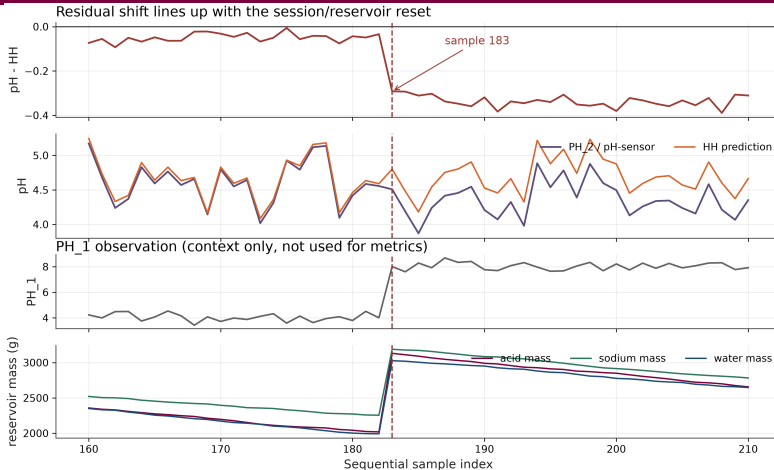
Data



Model performance

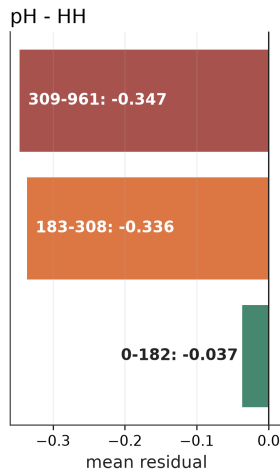
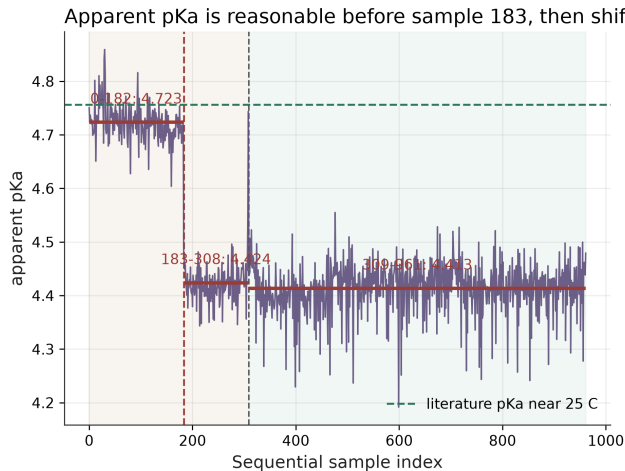


Residual shift evidence at sample 183



Basis for the session/overnight label: timestamp jumps from 2026-05-12 20:32 to 2026-05-13 13:37, the episode/step counters reset, and reservoir masses reset upward. PH_1 is shown only as context.

Apparent pKa confirms a regime-dependent intercept shift



$$pK_{a,app,k} = \text{pH}_k - \log_{10}(F_{\text{acetate},k}/F_{\text{acid},k})$$

Post-183 value is not a normal temperature pKa.

Raw three-flow control is non-unique for pH tracking

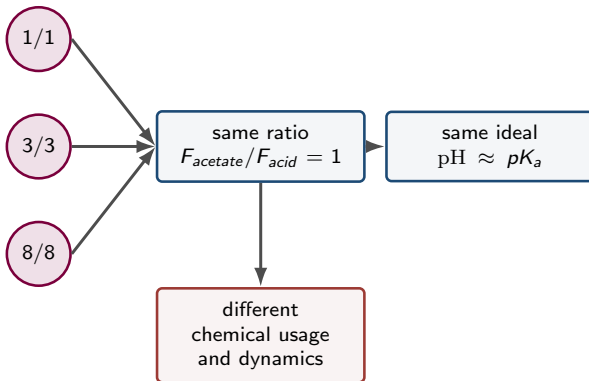
Problem with raw actions

$$u = \begin{bmatrix} F_{acid} \\ F_{acetate} \\ F_W \end{bmatrix}, \quad \text{pH} \approx pK_a + \log_{10} \frac{F_{acetate}}{F_{acid}}$$

$$(1, 1), (3, 3), (8, 8) \Rightarrow F_{acetate}/F_{acid} = 1$$

Document warning

- drifting to unnecessarily high flows,
- drifting to very low flows,
- unstable/chattering pump actions,
- meaningless use of water or total flow.



The pH reward alone cannot choose the “best” flow pair.

Controller structure 1: ratio-based controller

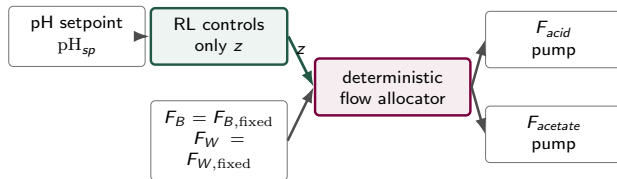
Reparameterize the action

$$r = \frac{F_{\text{acetate}}}{F_{\text{acid}}}, \quad z = \log_{10} \frac{F_{\text{acetate}}}{F_{\text{acid}}}$$

$$\text{pH} \approx \text{p}K_a + z, \quad F_B = F_{\text{acid}} + F_{\text{acetate}}$$

Deterministic flow allocator

$$r = 10^z, \quad F_{\text{acid}} = \frac{F_B}{1+r}, \quad F_{\text{acetate}} = \frac{rF_B}{1+r}$$



First RL version: RL controls only z ; F_B and F_W are fixed or chosen by a simple rule.

Controller structure 2: chemical usage objective

Chemical usage objective

$$\begin{aligned} R_k = & -Q(\text{pH}_k - \text{pH}_{sp})^2 - \lambda_{\Delta u} \|\Delta u_k\|^2 \\ & - \lambda_B (F_B - F_{B,ref})^2 - \lambda_{chem} (F_{acid} + F_{acetate}) \\ C_{buffer} \geq C_{min} \quad C_{buffer} = & \frac{C_0(F_{acid} + F_{acetate})}{F_{acid} + F_{acetate} + F_W} \end{aligned}$$

Clean hierarchy from the document

pH tracking	→	acid/acetate ratio
buffer strength	→	total acid+acetate flow
dilution/residence time	→	water or total flow

Pre-training implication

With this model and objective, we can start pre-training an RL agent, but we still have to decide the agent structure and the calibrated dynamic model used for training.